TOPIC:Student management system

LLD (Low-Level Design)

RAG-Based Student Management Assistant

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1. Modules

- Data Processing Module
- Chunking Module
- Embedding Module
- Vector Storage Module
- Retrieval Module
- Query Module
- LangGraph Module
- HITL Module

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2. Data Structures

Student Data:
```
{
  id: int,
  name: string,
  course: string,
  age: int
}
```

Query:
```
{
  query_text: string
}
```

Response:

```
{
  answer: string,
  confidence: float
}
```

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3. Workflow (LangGraph)

Input → Processing → Output

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4. Conditional Routing

- Low confidence → HITL
- Missing data → Escalation
- Valid answer → Output

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5. HITL Design

System detect kare:

- Answer unclear
- Data missing

Human ko send kare
Human response return

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6. API Design

Input:

```
{
  "query": "Show BCA students"
}
```

Output:

{
  "answer": "Rahul, Priya are in BCA"
}

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7. Error Handling

- No data → "No record found"
- Wrong query → fallback
- System error → retry